Assignment 5

ME/BME 736 — A Posteriori Error Estimation and Refinement

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Submission

The two notebooks in `assignments/assignment-05/notebooks/` contain working examples followed by the assignment tasks. Use the supplied code and extend it where requested; do not rebuild the finite element solvers. Include your calculations, plots, tables, and short explanations in the notebooks. No separate report is needed.

Submit `lastname_assignment05_gaussian.ipynb` and `lastname_assignment05_elasticity.ipynb`. Each must run without errors after **Restart Kernel and Run All Cells**. Follow `environment/README.md` and select **Python (Finite Elements)**. Run on one MPI process.

Problem 1 (60 marks): Residual-based refinement of a Gaussian solution

Notebook: `problem1.ipynb`. Reading: Chapter 8.

Compare residual-based element selection with selection based on the known elementwise error. Write the adaptive loop and show its indicators, marked elements, and refined meshes. Compare marking choices by the degrees of freedom needed to reach a specified true error. The three tasks carry 20, 25, and 15 marks.

Problem 2 (40 marks): Adaptive refinement of an elastic cantilever

Notebook: `problem2.ipynb`. Reading: the cantilever loading example in Chapter 8 and the linear-elastic formulation in Chapter 4.

Check the traction residuals and use them to select refinements for two fixed-width distributed loads. Show the selection and resulting meshes at three stages. Then use the supplied, separately identified point-force experiment to examine the loading model and the limits of the residual estimate. The three tasks carry 10, 20, and 10 marks.